

Adaptive Estimation of Pennes' Bio-Heat Equation

Guglielmo Cappellini*, Andrea Cristofaro*, Ernesto Staffetti[†], Giovanni Trappolini*, Marilena Vendittelli*

**Department of computer, control and management Engineering, Sapienza University of Rome, Rome, Italy*
 {cappellini,cristofaro,trappolini,vendittelli}@diag.uniroma1.it

[†]*Universidad Rey Juan Carlos, Madrid, Spain*
 ernesto.staffetti@urjc.es

Abstract—This paper reports on a novel approach proposed in [1], [2] for adaptive estimation of uncertain parabolic reaction-diffusion Partial Differential Equations (PDEs), specifically focusing on the Pennes' bio-heat Equation. The core concept involves observing heat transfer in biological tissues and achieving real-time temperature estimation within the interior of a spatial domain using boundary measurements. The study is motivated by the clinical need of improving the accuracy of hyperthermia treatments used in oncology to enhance the efficacy of chemo and radiotherapy. To this aim, the temperature estimation provided through the proposed method is meant to be used as feedback information in robot-assisted hyperthermia treatments. The proposed method demonstrates its effectiveness in estimating the system solution and accurately recovering the reaction coefficient. An innovative observer employing Neural Networks to predict PDE states eliminates the need for online integration, thus overcoming the limitations observed in classical numerical simulation tools. The preliminary results show the effectiveness of the method within the studied problem and its potential for extension to larger classes of problems and higher dimensions.

Index Terms—Adaptive estimation, Hyperthermia treatment, Multiple-Model Observer, Deep learning

I. INTRODUCTION

This study serves as a foundational stride toward the automatic delivery of hyperthermia treatments. Hyperthermia finds application in oncology to enhance the effectiveness of radiotherapy and chemotherapy [3], [4]. The treatment causes a temperature rise at the tumoral target through electro-magnetic radiation with appropriate frequency and power. Literature underscores that treatment success hinges upon achieving and maintaining target temperatures at appropriate timings and durations. Ensuring reproducibility across successive treatments is equally crucial [5], [6]. Accuracy in pre-operative plan transfer and radiating antenna positioning can be dramatically improved with the assistance of a robotic system¹. However, the absence of feedback control of the temperature at the target remains an obstruction to optimality of treatments.

The challenge lies in estimating the heat transfer phenomenon in the presence of uncertain coefficients and relying on, noninvasive, discrete, boundary measurements. This study proposes estimating the entire target volume's temperature via superficial measurements, enabling minimally invasive real-time temperature control. The challenge is preliminarily tackled within the 1D Pennes' bio-heat Equation. The estimation

relies on the multiple-model (MM) adaptive observer described in [1]. In the presented implementation the prediction step is performed by resorting to physics-informed deep learning techniques. To address the data scarcity issue, physics-informed learning incorporates physical laws into network training through the use of a set of physically-semantic losses and inducing a more robust learned space. In this work, we exploit the DeepXDE Python library allowing the solution of PDEs through Physics-Informed Neural Networks (PINNs) [7]. Beyond its practical utility, this approach paves the way for extending the observer to higher-dimensional domains.

A. 1D Pennes' bio-heat Equation

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, we denote by $\partial_{z_i} f$ the partial derivative of $f(z_1, \dots, z_n)$ with respect to the variable z_i , $i = 1, \dots, n$. The Pennes' Bio-heat Equation (PBHE) describes the evolution of the temperature $T(x, t)$ in a biological tissue, and it is given by the parabolic reaction-diffusion PDE

$$\rho c \partial_t T = \kappa_{\text{eff}} \partial_{xx} T - \rho_b \omega_b c_b (T - T_a) + \mathcal{Q} \quad (1)$$

where ρ is the tissue density, c is specific heat of the tissue, κ_{eff} is the thermal conductivity of the tissue, ρ_b is blood density, ω_b is the blood perfusion rate, c_b is the specific heat of blood, T_a is the arterial blood temperature considered constant at 37 °C [8] and $\mathcal{Q}$ is an internal heat source. The blood perfusion rate ω_b is bounded within a certain range, but the actual value is typically not known. The left-hand side of (1) expresses the variation of the thermal energy storage while, on the right-hand side, the first term is the net transfer of thermal energy due to temperature gradients, and the second is the blood perfusion term and its coefficient ω_b is uncertain.

Remark 1.1: Let us observe that, thanks to linearity, the dependency on the constant T_a can always be removed from equation (1) by defining the change of variables $\bar{T} := T - T_a$.

II. MULTIPLE-MODEL OBSERVER

In superficial and semi-deep Hyperthermia (up to 4 cm depth), when heating is provided by microwaves, three components can be identified: **1)** applicator to deliver power; **2)** Water bolus to improve the impedance matching and to keep the skin temperature comfortable; **3)** Thermocouples (TC) matrix to monitor skin temperature. In the following, we will consider a 1D geometry with total length $L_0 = 5\text{cm}$. This can be considered as a needle that enters the body perpendicularly

¹One of the authors has participated to the development of **ROBOHOT**, the first robotic system for superficial hyperthermia treatments.

with respect to the skin surface. The boundary in depth $x = 0$ is out of the heating range. The boundary $x = L_0$ represents the skin surface. For simplicity's sake, we will consider the thermal problem, i.e., how thermal power deposited by the applicator diffuses into the system. In reality, the power delivered by the hyperthermic system is of electromagnetic type. This, corrected for the reflected power, interacts with the biological tissues depending on their electromagnetic properties, and it is transformed into thermal power. It is usually defined as a specific absorption rate (SAR). The term $\mathcal{Q}$ in (1) comprehends the metabolic heat generation rate and the SAR [9].

A. scaled PBHE

To achieve robustness and stability of PINNs, it is recommended to scale input and outputs to $\mathcal{O}(1)$. We start by introducing the dimensionless temperature

$$\theta = \frac{T - T_a}{T_M - T_a},$$

where $T_M = 45^\circ\text{C}$ is assumed as the maximal tolerable skin temperature [10]. Also, we will consider the perfusion rate W defined as $W = \rho_b \cdot \omega_b$. In light of Remark 1.1 and upon appropriate transformations, detailed in [2] and inspired by [11], the 1D Pennes' Bio-heat Equation can be recast as a linear reaction-diffusion equation in the general form

$$a_1 \partial_\tau \theta = \partial_{XX} \theta - a_2 W \theta + a_3 \mathcal{Q} \quad (2)$$

defined for $(X, \tau) \in [0, 1] \times [0, +\infty)$, with Robin-type boundary conditions

$$\begin{aligned} \theta(0, \tau) &= C_0 \quad t \geq 0 \\ \partial_X \theta(1, \tau) &= v(\tau) \quad t \geq 0 \end{aligned} \quad (3)$$

where $C_0 \in \mathbb{R}$ is a known constant and $v(\tau)$ is a boundary input, and with initial condition

$$\theta(X, 0) = \theta_0(X) \quad X \in [0, 1]. \quad (4)$$

Assume that the value of the input $v(\tau)$ is accessible and that, in addition, the output

$$y(\tau) = \theta(1, \tau) \quad (5)$$

is available at any $\tau \geq 0$. Take further the following assumption, which will be used in the formulation of later results.

Assumption 2.1: The non-negative coefficient W satisfies:

i) W is constant; **ii)** there exist known lower and upper bound $W_{\min}, W_{\max}$ with $0 \leq W_{\min} \leq W \leq W_{\max}$.

Bearing this in mind, the main goal is to reconstruct asymptotically the solution $\theta(X, \tau)$ to the PDE (2) with boundary/initial conditions (3)-(4) using only measurements of $y(\tau)$ and knowledge of $v(\tau)$ and $\mathcal{Q}(X, \tau)$. With reference to the three components defined at the beginning of this section: **1)** $\mathcal{Q}(X, \tau)$ is the power deposited by the applicator; **2)** $v(\tau)$ is the heat removed by water bolus; **3)** $y(\tau)$ is the temperature measurement at the boundary, provided by temperature sensors.

B. Observer structure

Let us preliminarily address the observer design problem for the case of W fulfilling Assumption 2.1 and known. To construct an observer $\hat{\theta}(X, \tau)$, consider first a copy of the PDE dynamics (2)

$$a_1 \partial_\tau \hat{\theta} = \partial_{XX} \hat{\theta} - a_2 W \hat{\theta} + a_3 \mathcal{Q} \quad (6)$$

defined for $(X, \tau) \in [0, 1] \times [0, +\infty)$ with boundary conditions

$$\begin{aligned} \hat{\theta}(0, \tau) &= C_0 \\ \partial_X \hat{\theta}(1, \tau) &= v(\tau) + \alpha(y(\tau) - \hat{\theta}(1, \tau)) \end{aligned} \quad (7)$$

where $\alpha > 0$ is the output injection gain. The compatibility conditions for the initialization of the observer $\hat{\theta}_0(X)$ read then as

$$\hat{\theta}_0(0) = \theta_0(0) \wedge \partial_X \hat{\theta}_0(X)|_{X=1} + \alpha \hat{\theta}_0(1) = v(0) + \alpha \theta_0(1)$$

C. Multiple-model adaptive scheme

In the case of uncertain, constant W in a bounded admissible range, a family of observers is employed. Let us consider a set of fixed values $W_j \geq 0$, $j = 1, \dots, N$ for some $N \in \mathbb{N}$, with $W_1 < W_2 < \dots < W_N$ and $W_1 \leq W_{\min}$, $W_{\max} \leq W_N$. Accordingly, let us define a family of observers $\hat{\theta}^{(j)}(x, t)$ with the structure (6) and with the choice $\tilde{W} = W_j$, that is

$$\begin{aligned} a_1 \partial_\tau \hat{\theta}^{(j)} &= \partial_{XX} \hat{\theta}^{(j)} - a_2 W_j \hat{\theta}^{(j)} + a_3 \mathcal{Q} \\ \hat{\theta}^{(j)}(0, \tau) &= C_0 \\ \partial_X \hat{\theta}^{(j)}(1, \tau) &= v(\tau) + \alpha(y(\tau) - \hat{\theta}^{(j)}(1, \tau)). \end{aligned} \quad (8)$$

Following the approach proposed in [12], the idea is then to introduce an overall estimator

$$\hat{\theta}^\dagger(X, \tau) = p_1(\tau) \theta^{(1)}(X, \tau) + \dots + p_N(\tau) \theta^{(N)}(X, \tau) \quad (9)$$

obtained as dynamic convex combination of the observers $\theta^{(j)}(X, \tau)$ with weights $p_j(\tau)$ being updated according to

$$\dot{p}_j(\tau) = -\lambda \left(1 - \frac{e^{-\mu_j(\tau)}}{\sum_{\ell=1}^N p_\ell(\tau) e^{-\mu_\ell(\tau)}} \right) p_j(\tau) \quad j = 1, \dots, N \quad (10)$$

where $\mu_j(\tau) := v|\theta(1, \tau) - \hat{\theta}^{(j)}(1, \tau)|^2$, $v > 0$, is the (absolute) output error and $\lambda > 0$ is the adaptive gain.

The overall observer (9) corresponds to a weighted average of the individual observers, whose weights adapt based on the size of the associated output errors. Whenever one of these errors is ultimately smaller than the others, with

$$\lim_{\tau \rightarrow +\infty} \sup (\mu_{j^*}^*(\tau) - \mu_\ell(\tau)) < 0 \quad \forall \ell \neq j^*,$$

one may expect the convergence of the weights to the values

$$\lim_{\tau \rightarrow \infty} p_{j^*}(\tau) = 1, \quad \lim_{\tau \rightarrow \infty} p_\ell(\tau) = 0 \quad \forall \ell \neq j^*,$$

so that (9) reduces to $\hat{\theta}^\dagger(X, \tau) = p_{j^*}(\tau) \theta^{(j^*)}(X, \tau)$ (see [12, Theorem 1]). This means that $\hat{\theta}^{(j^*)}(X, \tau)$ results being the *best* observer among the considered family $\{\hat{\theta}^{(j)}(X, \tau)\}_{j=1, \dots, N}$, with respect to the inherent output error function.

III. NN-BASED OBSERVER IMPLEMENTATION

As described in Sect. II-C, the predictions of the observers are used to provide the estimation of the temperature (9), in the whole domain at each time instant τ , through a convex combination of weights with the dynamics (10). The novel application of Physics-Informed Neural Networks (PINNs) [13] to this setting, enabled the creation of a pipeline that is both effective in terms of accuracy, as highlighted by simulation results, and efficient in terms of overall computational time.

A. PINN for solving Pennes' equation

During the training, the network's weights are tuned to minimize the loss function $\mathcal{L}$, which measures the discrepancy between the neural network solution θ , the constraints represented by the PDE, and the boundary conditions on a set of sampling points in the spatiotemporal domain. We will name this network Neural Bio-Heat System (NBHS). NBHS will take the two coordinates (X, τ) as input, and will return its prediction for $\theta(X, \tau)$. Using a PINN, in this case, presents several advantages. In fact, this approach allows probing the system in a continuous manner and in real-time, meaning that the computation time is that of a forward pass of a deep neural network, which usually takes a few milliseconds on most hardware. Moreover, this approach can potentially scale to multiple input dimensions trivially. Alternatively, the use of a solver for each setting is necessary, with apparent drawbacks in terms of computational time.

B. Implementing a multiple-model observer with PINNs

The observer of the form (6), (7) is implemented by means of a second PINN, which we name Neural Bio-Heat Observer (NBHO). With respect to NBHS, NBHO needs the PINN architecture to be augmented with two extra inputs: the system's output $y(\tau)$ and known superficial flux $v(\tau)$. For the implementation of the multi-model estimator (9), each NBHO $_j$ will produce a specific prediction of the temperature on the boundary $\hat{\theta}^{(j)}(1, \tau)$. The error $\mu_j = |\theta(1, \tau) - \hat{\theta}^{(j)}(1, \tau)|$ is then used to compute the adaptive weights. The adaptive weights dynamics (10) is solved using the `solve_ivp` function in the `scipy` Python package.

C. Hyper-parameters optimization

Some issues regarding the adaptation of PINNs to our case study suggested a deeper analysis of hyper-parameters (HPs) tuning. This has been developed according to [14]. PINNs, like other neural network-based models, exhibit a number of HPs, such as the learning rate, the width and depth of the deep NN, the activation function, and the weights for the loss function. The high dimensionality for the HPs search space makes it difficult to find proper configurations. To remedy this concern, HPs optimization (HPO) is an insightful solution. While many approaches exist, guided by performance, we chose Gaussian processes-based Bayesian optimization.

Hyper-parameter (Symbol)	Range
Learning-rate (α)	[1.00E-05, 1.00E-01]
Depth (L-1)	[1, 8]
Width (N)	[5, 500]
Initialization (Φ_0)	Glorot uniform, He normal, He uniform
Activation function (σ)	relu, selu, silu, sigmoid, tanh
Loss weights (w)	[1, 250] $\in \mathbb{R}^4$

Each network is trained for $K = 20000$ iterations. The search space is Λ , the Cartesian product of all HP ranges. Every $\lambda \in \Lambda$ writes as: $\lambda = [\alpha, L - 1, N, \Phi_0, \sigma, w]$. The acquisition function allows to define the next point at each iteration. A common acquisition function is the negative expected improvement. The metric is the error of PINNs prediction with respect to MATLAB's exact solution. The starting hyper-parameter set is the same for both NBHS and NBHO: $\lambda_0 = [1e - 3, 4, 50, \text{"sin"}, [1, 1, 1, 1]]$. After 1000 iterations, the optimal configuration for the two networks is different. Typically, the improvement in prediction accuracy after HPO exceeds 200%.

IV. SIMULATIONS

With reference to [8], considering homogeneous muscle tissue: $a_1 = 1.061375$, $a_2 = 1.9125$, $a_3 = 6.25e - 05$, $W = 2.48 \frac{kg}{m^3s}$. The expression for Q has been formulated thanks to [15]: $Q = q_{met} + e^{-cL_0(X_0-s)}P$ with $q_{met} = 4200 \frac{W}{m^3}$, $c = 16 m^{-1}$, $X_0 = 0.09$, $P = 76142.131$. Initial and boundary conditions have been chosen as follows:

$$\begin{aligned}\theta(X, 0) &= 0 \quad \forall X \in [0, 1] \\ \theta(0, \tau) &= 0 \quad \forall \tau \geq 0 \\ \partial_x \theta(1, \tau) &= -1 \quad \forall \tau \geq 0\end{aligned}$$

Example 4.1: Let us assume that W is only known to belong to the interval $[W_{\min}, W_{\max}]$, and consider the multiple-model approach synthesized in Sect. II-C, with $N = 8$ models computed through a uniform gridding $\{W_1, W_2, \dots, W_8\}$ of the admissible variation interval of W , so that the closest value to the actual perfusion rate is W_5 . Each model $\hat{\theta}^{(j)}(X, \tau)$, $j = 1, 2, \dots, 8$ has output injection gain $\alpha = 1$ and initial condition:

$$\hat{\theta}(X, 0) = X \left(\frac{6}{5} - X \right)$$

The comparison of system solution $\theta(X, \tau)$, observers $\hat{\theta}^{(j)}(X, \tau)$ and MM adaptive observer $\hat{\theta}^\dagger(X, \tau)$ at the final time $t_f = 1$ is proposed in Figure 1. As clearly visible, the actual system is quite accurately reconstructed by the MM adaptive observer $\hat{\theta}^\dagger(X, \tau)$. The latter is built using the dynamic weights $p_j(X, \tau)$, whose time evolution is reported in Fig. 2 for $\lambda = 200$.

V. CONCLUSIONS

This paper reported a novel approach for adaptive estimation in the context of uncertain parabolic reaction-diffusion PDEs, with a focus on the Pennes' bio-heat equation. A multiple-model adaptive scheme, already proven to be efficient for

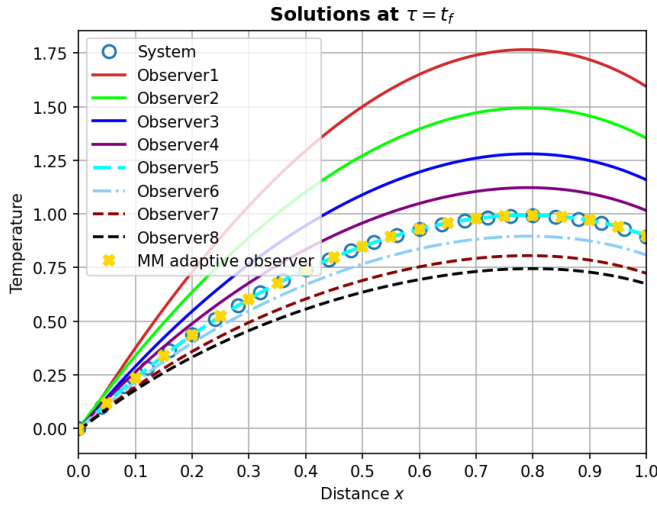


Fig. 1. Example 2: Solution vs. observers at $\tau = t_f$

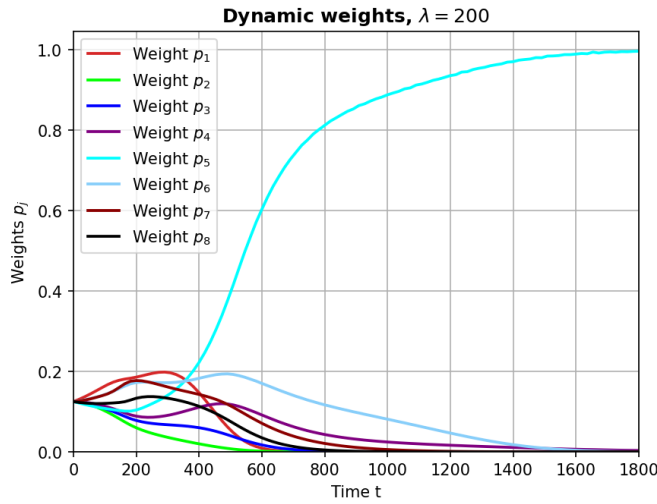


Fig. 2. Example 2: Dynamic weights, $\lambda = 200$

finite-dimensional systems, has been extended to the considered class of linear 1D PDEs with boundary outputs. The proposed multiple-model observer leverages deep learning techniques and PINNs to estimate the system solution and recover uncertain parameters. Simulation results indicate the potential applicability of the approach in biomedical hyperthermia treatments. Future work will be devoted to generalizing the observer design to higher dimensions and considering time-varying uncertain coefficients, as well as experimentation with phantom and clinical data.

VI. ACKNOWLEDGMENT

Supported by the ERC Advanced Grant 788893 AMDROMA, EC H2020RIA project “SoBigData++” (871042), PNRR MUR project PE0000013-FAIR”, PNRR MUR project IR0000013-SoBigData.it.

REFERENCES

- [1] Andrea Cristofaro, Guglielmo Cappellini, Ernesto Staffetti, Giovanni Trappolini, and Marilena Vendittelli. Adaptive Estimation of the Pennes’ Bio-Heat Equation - I: Observer design. In *accepted for presentation at the 62nd IEEE Conference on Decision and Control*, 2023.
- [2] Guglielmo Cappellini, Andrea Cristofaro, Ernesto Staffetti, Giovanni Trappolini, and Marilena Vendittelli. Adaptive Estimation of the Pennes’ Bio-Heat Equation - II: a NN-based implementation for real-time applications. In *accepted for presentation at the 62nd IEEE Conference on Decision and Control*, 2023.
- [3] N.R. Datta, S. Gómez Ordóñez, U.S. Gaipal, M.M. Paulides, H. Crezee, J. Gellermann, D. Marder, E. Puric, and S. Bodis. Local hyperthermia combined with radiotherapy and/or chemotherapy: Recent advances and promises for the future. *Cancer Treatment Reviews*, 41(9):742–753, 2015.
- [4] Pernille B. Elming, Brita S. Sørensen, Arlene L. Oei, Nicolaas A.P. Franken, Johannes Crezee, Jens Overgaard, and Michael R. Horsman. Hyperthermia: The optimal treatment to overcome radiation resistant hypoxia. *Cancers*, 11(1), 2019.
- [5] Akke Bakker, Jacoba van der Zee, Geertjan van Tienhoven, H. Petra Kok, Coen R. N. Rasch, and Hans Crezee. Temperature and thermal dose during radiotherapy and hyperthermia for recurrent breast cancer are related to clinical outcome and thermal toxicity: A systematic review. *International Journal of Hyperthermia*, 36(1):1023–1038, 2019.
- [6] Takayuki Ohguri, Yoko Harima, Hajime Imada, Hideyuki Sakurai, Tatsuya Ohno, Yoshiyuki Hiraki, Koh Tuji, Masahiro Tanaka, and Hiromi Terashima. Relationships between thermal dose parameters and the efficacy of definitive chemoradiotherapy plus regional hyperthermia in the treatment of locally advanced cervical cancer: Data from a multicentre randomised clinical trial. *International Journal of Hyperthermia*, 34(4):461–468, 2018.
- [7] Lu Lu, Xuhui Meng, Zhiping Mao, and George Em Karniadakis. DeepXDE: A deep learning library for solving differential equations. *SIAM Review*, 63(1):208–228, 2021.
- [8] M. L. Van Der Gaag, M. De Bruijne, T. Samaras, J. Van Der Zee, and G. C. Van Rhoon. Development of a guideline for the water bolus temperature in superficial hyperthermia. *International Journal of Hyperthermia*, 22(8):637–656, 2006.
- [9] J. Lang, B. Erdmann, and M. Seebass. Impact of nonlinear heat transfer on temperature control in regional hyperthermia. *IEEE Transactions on Biomedical Engineering*, 46(9):1129–1138, 1999.
- [10] Hana Dobšiček Trefná, Hans Crezee, Manfred Schmidt, Dietmar Marder, Ulf Lamprecht, Michael Ehmann, Josefin Hartmann, Jacek Nadobny, Johanna Gellermann, Netteke van Holthe, Pirus Ghadjar, Nicoletta Lomax, Sultan Abdel-Rahman, Christoph Bert, Akke Bakker, Mark D. Hurwitz, Chris J. Diederich, Paul R. Stauffer, and Gerard C. van Rhoon. Quality assurance guidelines for superficial hyperthermia clinical trials: I. clinical requirements. *International Journal of Hyperthermia*, 33(4):471–482, 2017. PMID: 28049386.
- [11] Junnosuke Okajima, Shigenao Maruyama, Hiroki Takeda, and Atsuki Komiya. Dimensionless solutions and general characteristics of bioheat transfer during thermal therapy. *Journal of Thermal Biology*, 34(8):377–384, 2009.
- [12] Vahid Hassani, A. Pedro Aguiar, António M. Pascoal, and Michael Athans. Further results on plant parameter identification using continuous-time multiple-model adaptive estimators. In *Proceedings of the 48th IEEE Conference on Decision and Control (CDC) held jointly with 2009 28th Chinese Control Conference*, pages 7261–7266, 2009.
- [13] M. Raissi, P. Perdikaris, and G.E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- [14] Paul Escapil-Inchauspé and Gonzalo A. Ruz. Hyper-parameter tuning of physics-informed neural networks: Application to helmholtz problems, 2022.
- [15] C Guiot, E Madon, D Allegro, P G Piantà, B Baiotto, and P Gabriele. Perfusion and thermal field during hyperthermia. experimental measurements and modelling in recurrent breast cancer. *Physics in Medicine & Biology*, 43(10):2831, oct 1998.